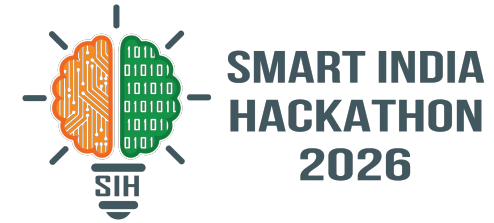


SMART INDIA HACKATHON 2026

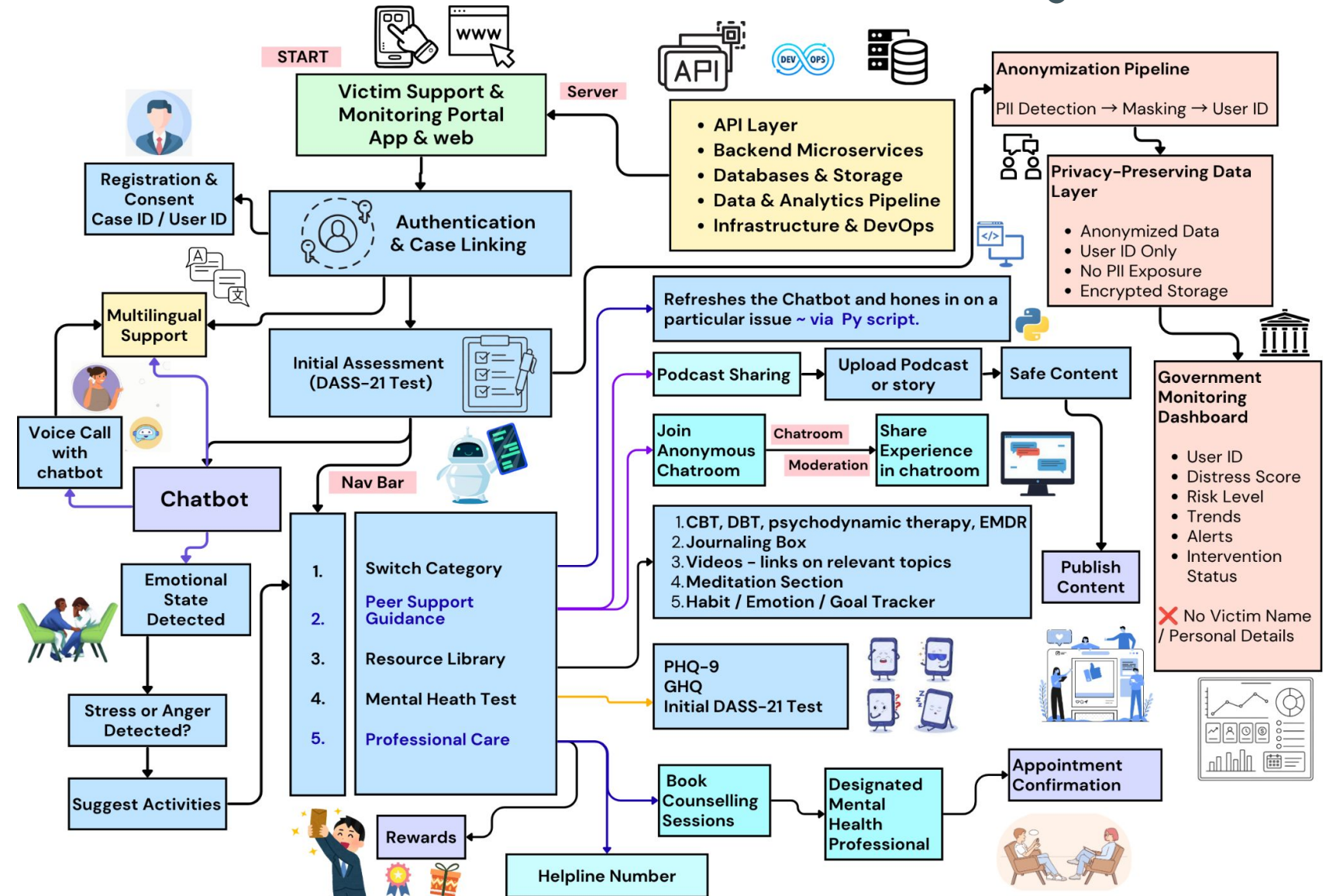
TITLE PAGE



- **Problem Statement ID** – 26094
- **Problem Statement Title-** AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-** shoonya_ek



- Existing systems mainly provide legal, financial, or basic support, but lack continuous AI-based mental health monitoring and distress prediction.
- Helplines and support channels often lack multilingual AI, voice-stress-analysis, behavioural monitoring, and automated risk detection.
- Mental health platforms do not monitor victims throughout investigation, trial, compensation, and rehabilitation.
- Counselling services provide human support, but lack dynamic distress scoring, predictive risk alerts, case-event correlation, and government-level intervention dashboards.



Backbone- Expo

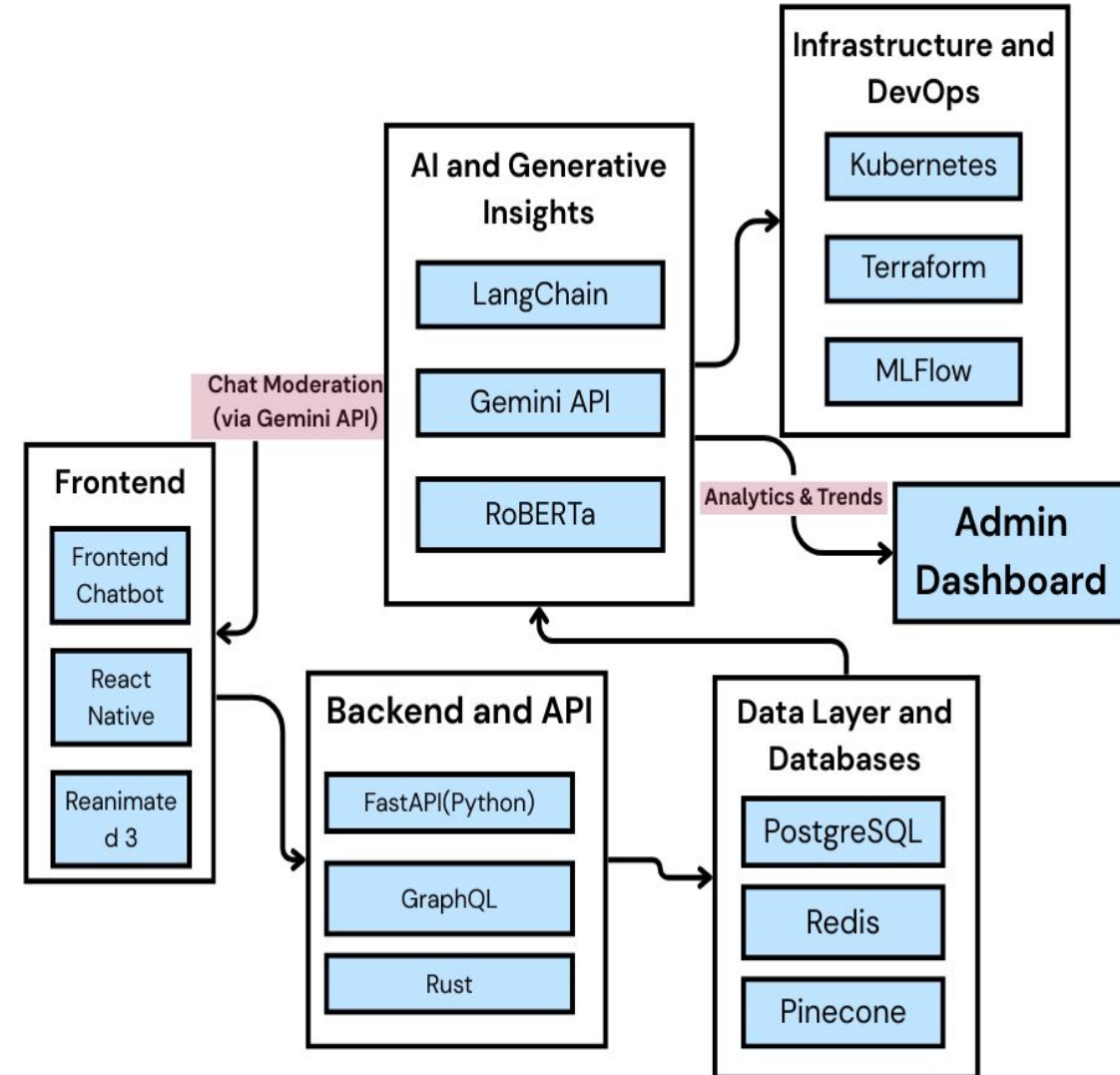
1.Frontend: React Native, Reanimated 3, Skia, Zustand

2.Backend & API: Python, FastAPI, spaCy

3.Data Layer & DB: PostgreSQL, MongoDB, Supabase, Redis, Vercel / Render

4.AIML & Gen. Insights: RoBERTa-based PII Detection, LangChain, Gemini API

5.Infrastructure & DevOps: Google Cloud Platform/ Amazon Web Services, MLFlow, Docker, Kubernetes, Terraform, Grafana





Feasibility:

1. Utilizes technologies like AI chatbots, cloud storage, and ML analytics.
2. Users are tech-friendly and more likely to use a chatbot than visit a counselor.
3. Primary costs - hosting, security certifications, counselor salaries.



Viability:

1. Revenue from institutional subscriptions and government funding.
2. Potential Users - Victims/Complainants, Counsellors, District & State Government Officials.
3. Scalability - Multiple districts, states, and national-level implementation.
4. Provides long-term value through predictive distress detection, timely alerts, and targeted victim support.



Challenges:

1. Technical Challenges - Accurate emotion detection, Data privacy & encryption.
2. Social Challenges - Trust-building, Counselor availability
3. Operational Challenges - Content moderation, Emergency response speed
4. Building trust among users to openly share mental health concerns.



Use cases:

1. AI chatbot for periodic victim check-ins, distress monitoring, and early-risk detection.
2. Dashboard to track anonymized mental health trends.
3. Podcasts for sharing personal stories anonymously.
4. Appointment management and real-time alerts.



Gov. Adoption Potential:

1. Government departments can adopt the platform for monitoring registered atrocity cases.
2. Integration with existing government portals, helplines, IVRS.
3. Partnerships with authorised counselling, healthcare and victim-support agencies.

Supporting Facts:

1. WHO reports 1 in 7 people aged 15-24 suffer from mental health conditions.
2. Suicide is the third leading cause of death among those aged 15–29 years old.



Feasibility:

1. Uses proven AI (NLP, Voice Analytics) and secure cloud architecture.
2. Accessible via low-tech IVRS and SMS.
3. Primary costs: Cloud hosting, PII encryption, and system integrations.



Viability:

1. B2G model funded directly by the Department of Social Justice.
2. Potential Users: Atrocity victims, designated counselors, and district/state nodal officers.
3. Scales easily across district, state, and national government dashboards.
4. Delivers long-term value via continuous tracking rather than passive helplines.



Challenges:

1. **Technical:** Ensuring zero PII leakage via strict anonymization pipelines.
2. **Social:** Building trust with traumatized victims using regional voice dialects.
3. **Operational:** Ensuring rapid on-ground intervention by authorities when the AI flags high-risk distress.



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IMPROVED MENTAL WELL-BEING

- Users receive timely emotional support.
- Early detection of distress.

INCREASED ACCESS TO COUNSELLING

- Easy booking for professional counselling sessions.
- Multilingual.

IMPACT ON TARGET AUDIENCE

ENGAGEMENT AND ROUTINE BUILDING

- Frequent interactive activities like breathing exercises.
- Timely notifications.

SAFE SPACE FOR EXPRESSION

- Users can share their thoughts and stories anonymously.
- Encourages open discussions.

Social Benefits

- **Personalized Support:** Chatbot adapts tone & content based on user assessments.
- **Data Driven Insights:** Gov. authorities get anonymized district, state, and national-level trends to improve victim support, rehabilitation, and intervention programs.
- **Threat to Police Alerts:** Automatically alerts the nearest police station with authorised victim details when threats or critical safety risks are detected.
-

Economic Benefits

- **Cost Prevention:** Early detection reduces expensive mental health treatments.
- **Scalable & Affordable:** Easily expands without heavy physical infrastructure.

Government & Operational Benefits

- **Predictive Case Prioritisation:** Flags high-risk cases before distress escalates.
- **Intervention Effectiveness:** Tracks which interventions improve victim well-being.
- **Systemic Risk Detection:** Identifies district/state-level patterns driving victim distress.

Feature	Elevana	Bright Sky	NHAA (14566)	My Support Pathway	Wysa
Dynamic Distress Scoring	✓	✗	✗	✗	✗
Voice & Speech Emotion Analytics	✓	✗	✗	✗	✗
Predictive Risk Escalation	✓	✗	✗	✗	✗
Multilingual IVRS / Phone Follow-ups	✓	✗	✓	✗	✗
Police & District Official Alerts	✓	✗	✗	✗	✗
Legal FIR & Docket Status Tracking	✓	✗	✓	✗	✗
Structured Trauma & Self-Help Pathways	✓	✓	✗	✓	✓
Incident / Evidence Journaling	✓	✓	✗	✗	✗
24/7 AI Chatbot Interaction	✓	✗	✗	✗	✓
Privacy & Encrypted Data	✓	✓	✓	✓	✓

Competitive Landscape Analysis:

<https://uk.bright-sky.org/>

<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1780979>

<https://www.innovatingmindsgroup.com/my-support-pathway>

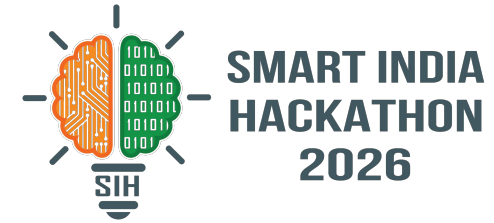
<https://www.wysa.com/>

References:

1. <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1780741>
2. <https://www.wysa.com/for-individuals>
3. <https://www.hestia.org/brightsky>
4. <https://www.who.int/campaigns/s-a-r-a-h>

Demo Link:

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.